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User manual

LXL Series:
Mini-EVA-Kit + ATTO-LXL



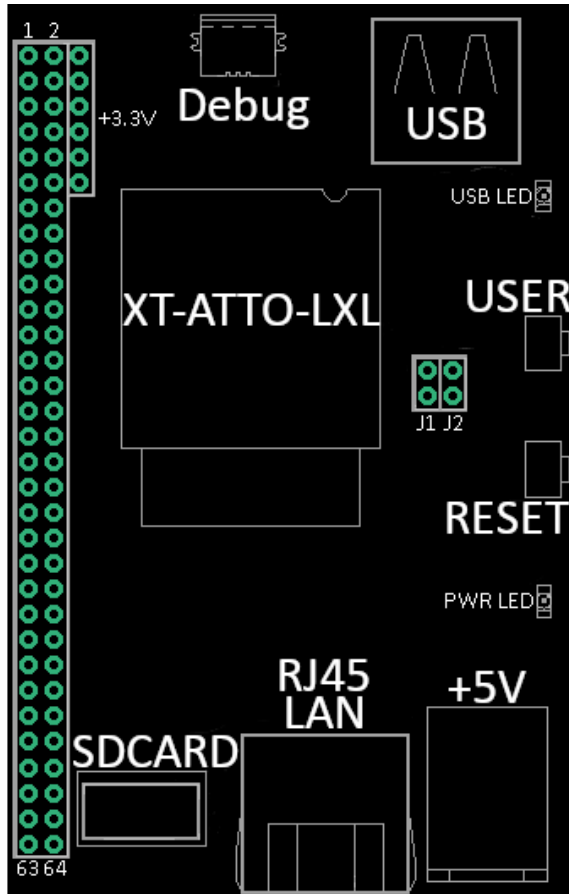
Contents

Hardware.....	3
Mini-EVA-Kit:.....	3
Power supply:.....	4
Debug USB:	4
Network:.....	4
Sdcard slot:.....	4
USB:	4
User button:	4
Reset button:	4
LED:	4
Pins:.....	4
Pins without interface configuration	5
Interface 1	5
Interface 2	6
Interface 2(continued)	7
Interface 3	7
Interface 4	8
Testtools/Testskripts AK-Nord:	8
TTL output (subfolder <i>ttlout/</i>):	8
TTL input (subfolder <i>ttlin/</i>):	8
ADC pins (subfolder <i>adc/</i>):	8
Software.....	9
SSH user:	9
Configuration of the custom models:	9
WEB server:	9
Python3:	9



Hardware

Mini-EVA-Kit:



Power supply +5V DC (+5V)
Network RJ45-Jack(RJ45 LAN)
Sdcard slot(SDCARD)
XT-ATTO-LXL Module
Debug-USB-C-Jack(Debug, COM-port settings: 115200 8 N 1)
USB-A Jack(USB)
USER button (USER)
RESET button(RESET)
PWR LED
USB LED
USB PWR-EN(J1)
USB OVCUR(J2)
Pin strip 6x pins (+3.3V)
Pin strip 64x pins (1-64)

**Power supply:**

+5V DC Power supply unit

Debug USB:

If you connect the EVA-Kit via a USB cable e.g. to a Windows PC, the operating system recognizes a serial interface like COM10. This can be operated via tools like putty with the following settings:
baudrate 115200, databits 8, parity N, stopbit 1

Network:

Please use the RJ45 LAN jack for the network connection

Sdcard slot:

Can be enabled via hardware configuration

USB:

USB-A socket to connect USB devices e.g. for Virtual-USB use.
To use USB, jumpers J1 and J2 must be set.

User button:

Is used to start the rescue mode at boot time. During operation it can be used for other functions. Pin: C28.

Reset button:

Restarts the XT-ATTO-LXL module

LED:

PWR LED indicates if power supply +5V is connected
USB LED indicates if USB-A socket is supplied with power +5V

Pins:

6x or 12x Pins(+3.3V): this can be used to supply small external sensors with 3.3V or to simulate different HW versions
2 rows of 32 pins each see the following interface tables.

You can configure the required interfaces as required under HARDWARE MENU->INTERFACEx. After a reboot of the ATTO module, the configured interfaces become active.



Pins without interface configuration

PIN	GPIO PIN	Beschreibung
1	GND	Stromversorgung GND
3	GND	Stromversorgung GND
5	GND	Stromversorgung GND
7	GND	Stromversorgung GND
9	GND	Stromversorgung GND
11	GND	Stromversorgung GND
43	+5V	+5V
44	PC28	Rescuemodus Button/Userbutton
45	WAKEUP	WAKEUP
46	Reset	Reset Resetbutton

Interface 1

PIN	GPIO PIN	Version 1_1	QSPI1 1_2	PWM 1_3
10	PB5	V4	QSPI1_SCK	PWMH2
8	PB6	V3	QSPI1_CS	PWML2
2	PB7	V0	QSPI1_IO0	PWMH3
4	PB8	V1	QSPI1_IO1	PWML3
6	PB9	V2	QSPI1_IO2	PWMFI1
12	PB10	V5	QSPI1_IO3	PWMEXTRG2



Interface 2

PIN	GPIO PIN	LCD 2_0	ISC 2_1	SPI1 2_3	PDMIC 2_4	UART3 2_5	QSPI1 2_6	I2SC1 2_7	FC3 USART 2_8u	FC3 SPI 2_8s	FC3 I2C 2_8i
13	PB11	DAT0			DAT	URXD3					
14	PB12	DAT1			CLK	UTXD3					
15	PB13	DAT2									
16	PB14	DAT3					SCK	MCK			
17	PB15	DAT4					CS	CK			
18	PB16	DAT5					IO0	WS			
19	PB17	DAT6					IO1	DI0			
20	PB18	DAT7					IO2	DO0			
21	PB19	DAT8					IO3				
22	PB20	DAT9									
23	PB21	DAT10							SCK	SPCK	
24	PB22	DAT11							RXD	MISO	TWCK
25	PB23	DAT12							TXD	MOSI	TWD
26	PB24	DAT13	D10						CTS	NPCS0	
27	PB25	DAT14	D11						RTS	NPCS1	
28	PB26	DAT15	D0								
29	PB27	DAT16	D1								
30	PB28	DAT17	D2								
31	PB29	DAT18	D3								
32	PB30	DAT19	D4								
33	PB31	DAT20	D5								
34	PC0	DAT21	D6								
35	PC1	DAT22	D7	SCK							
36	PC2	DAT23	D8	MOSI							
37	PC3	PWM	D9	MISO							
38	PC4	DISP	PCK	NPCS0							
39	PC5	VSYN	VSYN	NPCS1							
40	PC6	HSYN	HSYN	NPCS2							
41	PC7	PCK	MCK	NPCS3							
42	PC8	DEN	FIELD								



Interface 2(continued)

PIN	GPIO PIN	PDMIC 2_9	UART0 2_10	FC0 USART 2_11u	FC0 SPI 2_11s	FC0 I2C 2_11i	I2C0 2_12	CAN0 2_13	I2SC0 2_14
13	PB11								
14	PB12								
15	PB13								
16	PB14								
17	PB15								
18	PB16								
19	PB17								
20	PB18								
21	PB19								
22	PB20								
23	PB21								
24	PB22								
25	PB23								
26	PB24								
27	PB25								
28	PB26	DAT	URXD0						
29	PB27	CLK	UTXD0						
30	PB28			TXD	MOSI	TWD			
31	PB29			RXD	MISO	TWCK			
32	PB30			SCK	SPCK				
33	PB31			CTS	NPCS0		TWD0		
34	PC0			RTS	NPCS1		TWCK0		
35	PC1							CANTX0	I2SC0_CK
36	PC2							CANRX0	I2SC0_MCK
37	PC3								I2SC0_WS
38	PC4								I2SC0_DI0
39	PC5								I2SC0_DO0
40	PC6								
41	PC7								
42	PC8								

Interface 3

PIN	GPIO PIN	SPI1 3_1	FC2- SUART 3_2u	FC2- SPI 3_2s	FC2- I2C 3_2i	UART2 RX/TX 3_3	UART2 7PIN 3_3f	I2C0 3_4	ADC 3_5
47	PD23					URXD2	RXD		AD4
48	PD24					UTXD2	TXD		AD5
49	PD25	SPI1_SPCK					RTS		AD6
50	PD26	SPI1_MOSI	TX	MOSI	TWD		CTS		AD7
51	PD27	SPI1_MISO	RX	MISO	TWCK		DTR		AD8
52	PD28	SPI1_NPCS0	SCK	SPCK			DSR		AD9
53	PD29	SPI1_NPCS1	CTS	NPCS0			DCD	TWD0	AD10
54	PD30	SPI1_NPCS2	RTS	NPCS1				TWCK0	AD11



Interface 4

PIN	GPIO PIN	SDMMC1 4_1	WILC1000 SDIO 4_2
59	PA18	SDMMC1_DAT0	SDIO Data0
60	PA19	SDMMC1_DAT1	SDIO Data1
63	PA20	SDMMC1_DAT2	SDIO Data2
61	PA21	SDMMC1_DAT3	SDIO Data3
57	PA22	SDMMC1_CK	SD CLK
62	PA28	SDMMC1_CMD	SD CMD
64	PA30	SDMMC1_CD	

Testtools/Testskripts AK-Nord:

To use the test programs, you can log in to the ATTO module via ssh with the user admin(see section Software).

Path: `/home/admin/tools/`

TTL output (subfolder *ttlout/*):

With this individual pins can be tested as output. Here an example with the pin B12. After the call, B12 is set to HIGH for 2 seconds and then back to LOW again:

`./ttlouttest B 12`

TTL input (subfolder *ttlin/*):

This can be used to test the individual pins as input. Here an example with the pin B12. When called, B12 is read and at HIGH 1 is output, at LOW 0.:

`./ttlintest B 12`

ADC pins (subfolder *adc/*):

If you activate Interface 3_5, PINS PD23-PD30 can be used for ADC.

The following id's are valid for the test script:

PD23: 4
 PD24: 5
 PD25: 6
 PD26: 7
 PD27: 8
 PD28: 9
 PD29: 10
 PD30: 11

To test them you can apply DC voltage 0 to +3.3 volts to the pins e.g. connect directly to the EVA kit with the 3.3V pin row and execute the following script:

`./adcmvolt.sh 9`

You will get back the value in millivolts as result for the pin specified as parameter.



Software

SSH user:

Username: *admin*

Password: *ak-nord* (Please change necessarily)

Home folder: */home/admin/*

Configuration of the custom models:

There is a possibility to configure the model name of your products depending on version pins(PB6-PB10).

Configuration file: */home/admin/config/custom.ini*

Example:

[HARDWARE]

MANUFACTURER="COMPANY X"

CLASS="DEVICE SERVER"

[HW_MODEL]

111111="UARTBOX"

000010="USBBRIDGE"

WEB server:

Lighttpd is pre-installed and is started at boot time.

Root folder: */www/pages/*

Configuration file: */etc/lighttpd/lighttpd.conf*

Python3:

Python3 is installed on the ATTO with base package.

Additionally the package mpio from Microchip was installed:

<https://github.com/linux4sam/mpio>